AICC
25.300.1
CC-FS Interface Reference (RESTful)

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HUAWEI

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1 About This Document

Overview

This document describes the RESTful interfaces provided by the CC-FS. These interfaces are used for generating, storing, and downloading files of the call center platform and can support customer customization. This document also provides the usage instructions of these RESTful interfaces for reference by third-party business customization personnel.

Intended Audience

This document is intended for third-party business customization personnel.

- Call center business developers

Privacy Statement

This product provides secondary development capabilities and some of the interfaces are involved in processing customers' personal data (such as mobile numbers). You are allowed to provide or develop corresponding functions based on the business requirements and within the purpose and scope permitted by laws and regulations. During the customization development, you are obligated to take considerable measures to ensure that the personal data of customers is fully protected. The measures include but are not limited to the following:

- Collect and process individual data only for required businesses.
- Do not provide hidden channels for accessing the system without being authenticated based on normal mechanisms.
- Do not record sensitive data that may cause economic loss to customers, such as user password (plaintext or ciphertext) and bank account, in logs.

2 Change History

Issue 19 (2025-09-30)

This issue is released with AICC 25.300.0 and has no changes.

Issue 18 (2025-06-30)

This issue is released with AICC 25.200.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	Definitions of CDR and Recording Index Files Definition of Index Files for Agent Operation Details	Modified	Description specifying that the **service_no** field is meaningless is added.

Issue 17 (2025-03-31)

This issue is released with AICC 25.100.0 and has no changes.

Issue 16 (2025-01-15)

This issue is released with AICC 24.400.0 and has no changes.

Issue 15 (2024-09-30)

This issue is released with AICC 24.300.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	Description of Call Types (Recording Index)	New	Description of call types defined in recording index files is added.

Issue 14 (2024-06-30)

This issue is released with AICC 24.200.0 and has no changes.

Issue 13 (2024-03-30)

This issue is released with AICC 24.100.0 and has no changes.

Issue 12 (2024-01-30)

This issue is released with AICC 23.400.0 and has no changes.

Issue 11 (2023-11-15)

This issue is released with AICC 23.300.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://*ip*:*port*/CCFS/resource/ccfs/queryBillData https://*ip*:*port*/CCFS/resource/ccfs/queryAgentOprInfoData https://*ip*:*port*/CCFS/resource/ccfs/requestPlayVoice	Modified	Response codes are added to **Example**.

Issue 10 (2023-04-30)

This issue is released with AICC 23.100.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://*ip*:*port*/CCFS/resource/ccfs/queryBillData https://*ip*:*port*/CCFS/resource/ccfs/downloadBillFile	Modified	The following parameters are added to the CDR and recording index files downloaded using the interface for generating CDRs and recording indexes and the interface for downloading CDRs and recording indexes. Parameters added to CDR index files: **pre_device_in**: description of the previous device **leave_reason**: reason why a call is disconnected from a device **service_no**: type of the business currently provided by a device **device_in**: description of the current device **media_type**: media type of a call **sub_media_type**: submedia type

Issue 09 (2023-01-30)

This issue is released with AICC 22.200.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://*ip*:*port*/CCFS/resource/ccfs/downloadOiapRecord?locationId=xx	Modified	For the interface for downloading an intelligent IVR recording file, the **locationId** field in the request URL is changed from mandatory in CTI pool mode to optional.
2	https://*ip*:*port*/CCFS/resource/ccfs/downloadIVRRecordFile?locationId=xx	Modified	For the interface for downloading IVR voice messages, the **locationId** field in the request URL is changed from mandatory in CTI pool mode to optional.
3	https://*ip*:*port*/CCFS/resource/ccfs/queryBillData https://*ip*:*port*/CCFS/resource/ccfs/queryAgentOprInfoData	Modified	The location for storing uploaded third-party

No.	Interface	Type	Remarks
			certificates is modified.

Issue 08 (2022-06-30)

This issue is released with AICC 22.100.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://*ip*:*port*/CCFS/resource/ccfs/getRecordFileUrlFromObs	New	Interface for obtaining a recording download and playback URL.
2	https://*ip*:*port*/CCFS/resource/ccfs/queryBillData https://*ip*:*port*/CCFS/resource/ccfs/downloadBillFile	Modified	The following parameters are added to the CDR and recording index files downloaded using the interface for generating CDRs and recording indexes and the interface for downloading CDRs and recording indexes. Parameters added to CDR index files: **vdn**: VCC ID **pre_device_type**: type of the previous device that a call passes through **pre_device_no**: No. of the previous device that a call passes through **skill_id**: skill queue ID to which a call belongs **current_skill_id**: ID of the skill queue that processes a call Parameters added to recording index files: **vdn**: VCC ID **call_type**: call type **media_type**: media type **user_wanted_skill_id**: direction skill **current_skill_id**: ID of the skill queue that processes a call

Issue 07 (2022-01-30)

This issue is released with AICC 8.17.0 and has no changes.

Issue 06 (2021-10-15)

This issue is released with AICC 8.16.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://*ip*:*port*/CCFS/resource/ccfs/ivr/downloadFile	New	Interface used by a third party to download recording files.
2	https://*ip*:*port*/CCFS/resource/ccfs/ivr/uploadFile?vdnId=xxxx	New	Interface used by a third party to upload announcement files.
3	https://*ip*:*port*/CCFS/resource/ccfs/ivr/getFileHash	New	Interface used by a third party to generate the hash values in announcement files.
4	https://*ip*:*port*/CCFS/resource/ccfs/ivr/deleteFile	New	Interface used by a third party to delete uploaded announcement files.
5	https://*ip*:*port*/CCFS/resource/ccfs/downloadRecordFile	Modified	The format of the **fileName** field is modified.

Issue 05 (2021-08-23)

This issue is released with AICC 8.15.1 and has no changes.

Issue 04 (2021-07-20)

This issue is released with AICC 8.15.0 and has no changes.

Issue 03 (2021-03-31)

This issue is released with AICC 8.14.0 and has no changes.

Issue 02 (2020-12-30)

This issue is released with AICC 8.13.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://*ip*:*port*/CCFS/resource/ccfs/downloadIVRRecordFile?locationId=xx	New	Interface for downloading IVR messages.
2	https://*ip*:*port*/CCFS/resource/ccfs/downloadOiapRecord?locationId=xx	New	Interface for downloading intelligent IVR voice recording files.

Issue 01 (2020-09-30)

This issue is the first official release.

3 Overview

3.1 Function Description

The CC-FS component of the call center provides RESTful interfaces. The businesses can encapsulate and invoke the interfaces to generate, store, and download files, including recording playback customized by customers.

3.2 Interface Parameter Usage Principle

When using the RESTful interfaces provided by the CC-FS, the third-party business side must follow the usage principles described in Table 1.

Table 1 Interface parameter usage principle

Type	Description
Mandatory	Mandatory parameters must be contained in interface invoking requests sent from the business side to the CC-FS in any case.
Optional	Whether to use optional parameters can be determined based on business requirements.

3.3 Interface Encoding Description

The interfaces provided by the CC-FS use HTTPs+JSON for encoding. The **Content-Type: application/json; charset=UTF-8** message header information must be configured in HTTPs request.

4 Interface Description

NOTE

The NSLB server IP address involved in the request URLs of all interfaces in the following sections is the IP address of the outer NSLB server, and the port number is the mount port.

4.1 Data Access Interface

4.1.1 Obtaining a Recording Download and Playback URL

Description

This interface is invoked to upload a recording file on the CC-FS to OBS/LSS and obtain the recording download and playback URL from OBS/LSS.

- Prerequisites: Recording files have been synchronized to the CC-FS database by a scheduled task.
- Usage restrictions
  Developers can download only recordings under their own accounts. The developer account (**ak**) must correspond to the recording (**callId**).

The developer account (**ak**) is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

callId, **beginTime**, and **endTime** uniquely identify a recording.

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/getRecordFileUrlFromObs

Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

Parameter	Mandatory	Value Type	Default Value	Description
Content-Type	Yes	String	None	The value is fixed to **application/json; charset=UTF-8**.
Authorization	Yes	String	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

Parameter	Mandatory	Value Type	Default Value	Description
callId	Yes	String (1–25)	None	Call ID.
beginTime	Yes	String	None	Recording start time. The time is in *yyyy-MM-dd HH:mm:ss* format.
endTime	Yes	String	None	Recording end time. The time is in *yyyy-MM-dd HH:mm:ss* format. The interval between the recording start time and end time cannot exceed three days.
version	No	String	None	Current interface version, which is 2.0. **NOTICE:**

Table 2 Parameters in the request body

Parameter	Mandatory	Value Type	Default Value	Description
				Time zones other than GMT+8: **beginTime** and **endTime** must be set to time in the time zone where the tenant space is located, and **version** is mandatory. GMT+8 When **beginTime** and **endTime** are set to time in the GMT+0 time zone, **version** is optional. For example, if the recording start time is 2022-01-02 10:00:00 in the GMT+8 time zone, you can set **beginTime** to 2022-01-02 02:00:00. When **beginTime** and **endTime** are set to time in the time zone where the tenant space is located, **version** is mandatory.

Response

Table 3 Parameters in the response

Parameter	Value Type	Description
resultCode	String	Result code returned. **0**: success Other values: failure
resultDesc	String	Request result description. For details, see Error Code Reference .
resultData	Object	Response data.
url	String	Recording download and playback URL returned from the OBS/LSS after this interface is successfully invoked. The URL expires after 8 hours by default. If a call ID corresponds to multiple recording files, the download and playback URLs of multiple recording files are returned. The URLs are sorted by start time and end time.

Error Codes

For details, see Error Code Reference .

Example

- Request header

```
POST /CCFS/resource/ccfs/getRecordFileUrlFromObs HTTP/1.1
Authorization: ************
Accept: */*
```

```
Host: 10.154.198.164
Content-Type: application/json;charset=UTF-8
Content-Length: 185
```

- Request parameters


```
{
    "callId": "1637742300-27",
    "beginTime": "2021-11-24 06:25:03",
    "endTime": "2021-11-25 06:25:11",
    "version": "2.0"
}
```


- Response header

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
Date: Mon, 02 Jul 2018 02:43:03 GMT
```

- Response parameters


```
{
    "resultData": {
        "url": "https://south-aicc-develop.obs.cn-south-1.myhuaweicloud.com:443/ccfs/record/developId/13ddcde6-2072-40d2-b96f-58cac16e18f5?AccessKeyId=************&Expires=1637936620&Signature=************"
    },
    "resultCode": "0",
    "resultDesc": "success"
}
```


4.1.2 Generating CDRs and Recording Indexes

Description

This interface is invoked to generate CDR files and recording index files based on the specified conditions. After a file is generated, the file name (**billFileName**) is sent to the callback URL specified by **callBackURL** in callback mode.

The callback of the URL specified by **callBackURL** requires authentication. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Usage Description

- Prerequisites
    - The developer environment has been established and calls can be made properly. The developer has an independent server and the server has been bound to the callback URL (**callBackURL**).

- **accountId**, **agentId**, and **callId** are optional parameters, which can be flexibly combined to generate CDRs and recording index files. If none of the three parameters is specified, the corresponding data records of the developer in the specified period are generated.
- You have passed the authorization and obtained the access URL.
- You have set **aicc.ssl.trustAll** in the CC-FS configuration file **/home/ccfsapp/webapps/ccfsapp/WEB-INF/classes/config/servicecloud.base.properties** is to **false** and performed the following steps to place the certificate:

  1. Log in to the CC-FS server as an O&M user and switch to the **root** user.
     ```
     su - root
     ```
  2. Upload the third-party certificate that needs to be loaded by the callback URL for notifying the index file generated by the CC-FS to the same directory (**$HOME/conf**) as the **truststore.jks** certificate file of the AICC. The directory varies according to the actual environment.
  3. Change the owner group of the uploaded third-party certificate.
     ```
     chown ccfsapp:ccfsapp XXX.jks
     ```
  4. Import the third-party certificate to *truststore.jks*.
     ```
     keytool -import -alias xxx -file XXX.jks -keystore truststore.jks
     ```
     Enter the password of the *truststore.jks* certificate file as prompted.
  5. Delete the *XXX.jks* certificate from the environment.
  6. Restart the CC-FS.
     Go to the **/home/ccfsapp/bin** directory and run the following command:
     ```
     ./shutdown.sh;./startup.sh
     ```

- Usage restrictions
  A developer can access only the data of the account. The developer account (**appId**) must correspond to the enterprise account (tenant space ID: **accountId**).
  The developer account (**appId**) is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/queryBillData
Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameters in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	beginTime	String	Yes	None	Start time of CDRs and recordings, which is the time in the time zone where the tenant space is located. The interval between the start time and end time cannot exceed three days. The time is in *yyyy-MM-dd HH:mm:ss* format.
2	endTime	String	Yes	None	End time of CDRs and recordings, which is the time in the time zone where the tenant space is located. The interval between the start time and end time cannot exceed three days. The time is in *yyyy-MM-dd HH:mm:ss* format.
3	accountId	String (1–20)	No	None	Enterprise account (tenant space ID).
4	agentId	String (1–20)	No	None	Agent ID.
5	callId	String (1–25)	No	None	Call ID, which can be obtained from the file returned after the interface for downloading CDR and recording indexes is invoked.
6	callerNo	String (1–25)	No	None	Calling number.
7	calleeNo	String (1–25)	No	None	Called number. If **calleeNo2** does not exist, the value is the called number. If **calleeNo2** exists, the value is the calling number.

Table 3 msgBody parameters in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
8	calleeNo2	String (1–25)	No	None	Called number 2.
9	dataType	String (1–11)	Yes	None	Type of the data record file to be generated. The options are as follows: **call**: CDR **record**: recording index **call_record**: CDR and recording index
10	callBackURL	String	Yes	None	Callback URL. After generating a data record file, the platform returns **billFileName** to the URL specified by this parameter.

Response

The response obtained by invoking this interface consists of two parts:

(1) Response of the invoked party

(2) Response of the invoking part (response to third-party callback)

- Response of the invoked party
  Table 4 describes the parameters in the response.
- Response of the invoking party (response to third-party callback): The response consists of **request** and **msgBody**.
  request contains **version** (**2.0** by default).
  msgBody consists of **responseId** (part of the CDR file name), **billName** (name of the ZIP package of CDRs to be downloaded), and **accountId** (tenant space ID). Table 5 describes **resultData** in the response.

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. **0**: success Other values: failure
2	resultDesc	String	Request result description. For details, see Error Code Reference .
3	resultData	Object	Response data. For details, see Table 5.
4	response	Object	Request result object. For details, see Table 6.

Table 4 Parameters in the response

No.	Name	Value Type	Description
			This field is reserved for compatibility with earlier versions and is not recommended.

Table 5 resultData parameter in the response

No.	Name	Value Type	Description
3.1	responseId	String	Unique ID generated after the interface is successfully invoked. It is a part of the file name.

Table 6 response parameters in the response

No.	Name	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, 2.0.
4.2	resultCode	String (1–32)	Result code returned. For details, see Error Code Reference .
4.3	resultMsg	String	Request result description.

Table 7 msgBody parameters in the callback response body

No.	Name	Value Type	Description
1	responseId	String	Unique ID generated after the interface is successfully invoked. It is a part of the file name.
2	billFileName	String	Name of a CDR or recording index file. The value is in *yyyyMMdd_{responseId}*.**zip** format. For example, if the date when this interface is invoked is 2018-07-02 and the obtained value of **responseId** is **9239cb50-a838-4bb0-ab50-3441bf089446**, the value of this parameter is **20180702_9239cb50a8384bb0ab503441bf089446.zip**.
3	accountId	String	Enterprise account (tenant space ID). The enterprise account carried in the request parameter is used to associate the file name with an enterprise. If **accountId** is not set in the request parameter, this value is not returned.

Example

- Request header

```
POST /CCFS/resource/ccfs/queryBillData HTTP/1.1
Authorization: auth-v2/ak/2018-07-02T02:42:49Z/content-length;content-type;host
/7a8fb9d620ee488*****************85c5df0cce3c6a253
Accept: */*
Content-Type: application/json;charset=UTF-8
```

```
Content-Length: 297
```

- Request parameters


```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "beginTime": "2018-06-29 10:42:49",
        "endTime": "2018-07-02 10:42:49",
        "accountId": "",
        "agentId": "",
        "callId": "",
        "dataType": "call_record",
        "callBackURL": "https://10.57.118.171:8000"
    }
}
```


- Response header

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
Date: Mon, 02 Jul 2018 02:43:03 GMT
```

- Response


```
{
    "response": {
        "version": "2.0",
        "resultCode": "0",
        "resultMsg": "success"
    },
    "resultData": {
        "responseId": "9239cb50a8384bb0ab503441bf089446"
    },
    "resultCode": "0",
    "resultDesc": "success"
}
```


- Example of the message returned by the call center to the callback URL

  Message header:

```
POST / HTTP/1.1
Accept: */*
Content-Type: application/json;charset=UTF-8
Authorization: auth-v2/ak/2018-07-02T02:43:03Z/content-length;content-type;host
/c5bd683******************9def5747bd572c28
```

```
Host: 10.57.118.171
Content-Length: 208
```

Message parameters:


```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "responseId": "9239cb50a8384bb0ab503441bf089446",
        "billFileName": "20180702_9239cb50a8384bb0ab503441bf089446.zip",
        "accountId": ""
    }
}
```


4.1.3 Downloading CDRs and Recording Indexes

Description

This interface is invoked to download the .zip file (*yyyymmdd_ responseId*) of the CDR or recording package. You can view the CDR description (*yyyyMMddHHmmssSSS+ Three-digit random number*_**call.csv**) and the detailed indexes (*yyyyMMddHHmmssSSS+ Three-digit random number*_**record.csv**) generated for each recording.

Usage Description

- Prerequisites
  - CDRs and recording index files have been generated at the backend by invoking the interface for generating CDRs and recording indexes.
  - You have passed the authorization and obtained the access URL.
- Usage restrictions
  Developers can only download files under their own accounts. The developer account (**appId**) must correspond to the data record file name (**billFileName**).
  The developer account (**appId**) is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/downloadBillFile

Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	billFileName	String (1–49)	Yes	None	Name of a CDR and recording index file, which has been obtained by invoking the interface for generating CDRs and recording indexes. The value is in *yyyyMMdd_{responseId}*.**zip** format. For example, if the interface for generating CDRs and recording indexes has been invoked on **2018-07-02**, and if the obtained recording index **responseId** is **9239cb50a8384bb0ab503441bf08944**, the value of this parameter is **20180702_9239cb50a8384bb0ab503441bf089446.zip**.

Response

If the interface is successfully invoked, the system obtains the binary data of the file from the HTTP response message to generate a CDR file. The file name extension is .zip. For details about the parameters in the file, see Definitions of CDR and Recording Index Files . In the response message, the value of **content-type** is **Application/Octet-stream;charset=UTF-8**.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see Table 4. **resultData** is a reserved field and is left empty by default.

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.
4	response	Object	Request result object. For details, see Table 5. This field is reserved for compatibility with earlier versions and is not recommended.

Table 5 Parameters in a response message

No.	Name	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, 2.0.
4.2	resultCode	String (1–32)	Result code returned.
4.3	resultMsg	String	Request result description.

Example

- Request header

```
POST /CCFS/resource/ccfs/downloadBillFile HTTP/1.1
Authorization: auth-v2/ak/2018-07-02T02:43:08Z/content-length;content-type;host
/ae066c2f5d*****************13a0afc161cb7e66f5d
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "billFileName": "20180702_9239cb50a8384bb0ab503441bf089446.zip"
    }
}
```

4.1.4 Downloading a Recording File

Description

This interface is invoked to download a single recording file.

Usage Description

- Prerequisites
  - Recording index files have been obtained by invoking the interface for downloading CDRs and recording indexes.
  - You have passed the authorization and obtained the access URL.
- Usage restrictions
  Developers can only download files under their own accounts. The developer account (**appId**) must correspond to the recording file name (**fileName**).
  The developer account is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/downloadRecordFile
Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.
If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	fileName	String (1–225)	Yes	None	Relative path of the recording file, which is obtained from the **file_name** field in the recording index file returned when the interface for downloading CDRs and recording indexes is invoked. The value is in the */{nodeId}/{ccId}/record/{vdnId}/{yyyymmdd}/{agentid}/{original file name}*.**wav** format. Note: *{nodeId}* indicates the node ID, *{ccId}* indicates the call center ID, *{vdnId}* indicates the VDN ID, *{yyyymmdd}* indicates the date when the recording file is generated, *{agentid}* indicates the ID of the agent who generates the recording file, and *{original file name}* indicates the name of the recording file. Example: **/10/1/record/appId/100/15362/autoTest.wav**

Response

If the interface is successfully invoked, the system obtains the binary data of the file from the HTTP response message to generate a recording file. The file name extension is .wav. In the response message, the value of **content-type** is **Application/Octet-stream;charset=UTF-8**.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see Table 4. **resultData** is a reserved field and is left empty by default.

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.
4	response	Object	Request result object. For details, see Table 5. This field is reserved for compatibility with earlier versions and is not recommended.

Table 5 Parameters in a response message

No.	Name	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, 2.0.

Table 5 Parameters in a response message

No.	Name	Value Type	Description
4.2	resultCode	String (1–32)	Result code returned. For details, see Error Code Reference .
4.3	resultMsg	String	Request result description.

Example

- Request header

```
POST /CCFS/resource/ccfs/downloadRecordFile HTTP/1.1
Authorization: auth-v2/ak/2018-07-02T02:45:50Z/content-length;content-type;host/
eb453f68e85*****************80196c509c4913
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "fileName": "/10/1/record/appId/100/autoTest.wav"
    }
}
```

4.1.5 Downloading a Recording File (Extended)

Description

This interface is invoked to query and download a recording file by call ID or file name.
Intended audience: This interface is invoked by a third-party server and can be used in scenarios such as dumping.

Usage Description

- Prerequisites
  - Recording index files have been obtained by invoking the interface for downloading CDRs and recording indexes.
  - You have passed the authorization and obtained the access URL.

- Usage restrictions
  Developers can only download files under their own accounts. The developer account (**appId**) must correspond to the original recording file name (**fileName**).
  The developer account (**appId**) is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/downloadRecord
Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.
If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameters in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	fileName	String (1–225)	No	None	Recording file name, which is obtained from the **original_file_name** field in the recording index file returned when the interface for downloading CDRs and recording indexes is invoked. For example, the value is **X:/4/0/20180416/512/1629533.V3**. Note: Either **fileName** or **callId** must be set.

Table 3 msgBody parameters in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
2	callId	String (1–25)	No	None	Call ID. Note: At least one of **fileName** and **callId** must be set. If **callid** corresponds to multiple recording files, the .zip package will be downloaded.
3	ccId	String (1–64)	No	None	ID of a call center. The default value is **1**.

Response

If the interface is successfully invoked, the system obtains the binary data of the file from the HTTP response message to generate a recording file. The file name extension is .wav. If multiple recording files are downloaded by **callid**, the file name extension is .zip. In the response message, the value of **content-type** is **Application/Octet-stream;charset=UTF-8**.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see Table 4. **resultData** is a reserved field and is left empty by default.

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.
4	response	Object	Request result object. For details, see Table 5. This field is reserved for compatibility with earlier versions and is not recommended.

Table 5 Parameters in a response message

No.	Name	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, 2.0.
4.2	resultCode	String (1–32)	Result code returned. For details, see Error Code Reference .
4.3	resultMsg	String	Request result description.

Example

- Request header

```
POST /CCFS/resource/ccfs/downloadRecord HTTP/1.1
```

```
Authorization: auth-v2/ak/2018-07-02T02:44:42Z/content-length;content-type;host
/f18f6dd19****************6b8ff99f6c5884
Accept: */*
Host: 10.154.198.164
Content-Type: application/json;charset=UTF-8
Content-Length: 185
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "fileName": "Y:/autoTest.V3",
        "callId": "autoTest-1",
        "ccId": "1"
    }
}
```

4.1.6 Generating Indexes of Agent Operation Details

Description

This interface is invoked to generate an index file of agent operation details based on the conditions. After a file is generated, the file name (**agentOprInfoFileName**) is sent to the callback URL carried by **callBackURL** in callback mode.

The callback of the URL specified by **callBackURL** requires authentication. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Usage Description

- Prerequisites
  - The developer environment has been established and agents can sign in and perform operations properly. The developer has an independent server and the server has been bound to the callback URL (**callBackURL**).
  - The **accountId** and **agentId** can be freely combined to generate an index file for the agent operation details. If neither of the parameters is specified, the corresponding data records of the **appId** in the specified period are generated.
  - You have passed the authorization and obtained the access URL.
  - You have set **aicc.ssl.trustAll** in the CC-FS configuration file **/home/ccfsapp/webapps/ccfsapp/WEB-INF/classes/config/servicecloud.base.properties** is to **false** and performed the following steps to place the certificate:

1. Log in to the CC-FS server as an O&M user and switch to the **root** user.

```
su - root
```

2. Upload the third-party certificate that needs to be loaded by the callback URL for notifying the index file generated by the CC-FS to the same directory (**$HOME/conf**) as the **truststore.jks** certificate file of the AICC. The directory varies according to the actual environment.

3. Change the owner group of the uploaded third-party certificate.

```
chown ccfsapp:ccfsapp XXX.jks
```

4. Import the third-party certificate to *truststore.jks*.

```
keytool -import -alias xxx -file XXX.jks -keystore truststore.jks
```

   Enter the password of the *truststore.jks* certificate file as prompted.

5. Delete the *XXX.jks* certificate from the environment.

6. Restart the CC-FS.
   Go to the **/home/ccfsapp/bin** directory and run the following command:

```
./shutdown.sh;./startup.sh
```

- Usage restrictions
  A developer can access only the data of the account. The developer account (**appId**) must correspond to the enterprise account (**accountId**).
  The developer account is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/queryAgentOprInfoData

Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameters in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	beginTime	String	Yes	None	Start time of agent operations, which is the time in the time zone where the tenant space is located. The interval between the start time and end time cannot exceed three days. The time is in *yyyy-MM-dd HH:mm:ss* format.
2	endTime	String	Yes	None	End time of agent operations, which is the time in the time zone where the tenant space is located. The interval between the start time and end time cannot exceed three days. The time is in *yyyy-MM-dd HH:mm:ss* format.
3	accountId	String (1–20)	No	None	Enterprise account (tenant space ID).
4	agentId	String (1–20)	No	None	Agent ID.
5	callBackURL	String	Yes	None	Callback URL. After generating a data record file, the platform returns **fileName** to the URL specified by this parameter.

Response

The response obtained by invoking this interface consists of two parts:

(1) Response of the invoked party

(2) Response of the invoking part (response to third-party callback)

- Response of the invoked party

  Table 4 describes the parameters in the response.

- Response of the invoking party (response to third-party callback)

  The response consists **request** and **msgBody**. Table 7 describes **msgBody** parameters.

request contains **version** (**2.0** by default).

msgBody consists of **responseId** (part of the agent operation details file name), **fileName** (name of the ZIP package of agent operation details to be downloaded), and **accountId** (tenant space ID).

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. **0**: success Other values: For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	For the description about parameters in **resultData**, see Table 5.
4	response	Object	Request result object. For details, see Table 6. This field is reserved for compatibility with earlier versions and is not recommended.

Table 5 resultData parameter in the response

No.	Name	Value Type	Description
3.1	responseId	String	Unique ID generated after the interface is successfully invoked. It is a part of the file name.

Table 6 response parameters in the response

No.	Name	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, 2.0.
4.2	resultCode	String (1–32)	Result code returned. For details, see Error Code Reference .
4.3	resultMsg	String	Request result description.

Table 7 msgBody parameters in the callback response body

No.	Name	Value Type	Description
1	responseId	String	Unique ID generated after the interface is successfully invoked. It is a part of the file name.
2	fileName	String	Name of a CDR or recording index file. The value is in *yyyyMMdd_{responseId}*.**zip** format. For example, if the date when this interface is invoked is 2018-07-02 and the obtained value of **responseId** is **9239cb50a8384bb0ab503441bf089446**, the value of this parameter is **20180702_9239cb50a8384bb0ab503441bf089446.zip**.

Table 7 msgBody parameters in the callback response body

No.	Name	Value Type	Description
3	accountId	String	Enterprise account (tenant space ID). The enterprise account carried in the request parameter is used to associate the file name with an enterprise. If **accountId** is not set in the request parameter, this value is not returned.

Example

- Request header

```
POST /CCFS/resource/ccfs/queryAgentOprInfoData HTTP/1.1
Authorization: auth-v2/ak/2018-07-02T02:42:49Z/content-length;content-type;host
/7a8fb9d620e******************5c5df0cce3c6a253
Accept: */*
Host: 10.154.198.164
Content-Type: application/json;charset=UTF-8
Content-Length: 297
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "beginTime": "2018-06-29 10:42:49",
        "endTime": "2018-07-02 10:42:49",
        "accountId": "",
        "agentId": "",
        "callBackURL": "https://10.57.118.171:8000"
    }
}
```

- Response header

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
Date: Mon, 02 Jul 2018 02:43:03 GMT
```

- Response parameters

```
{
    "response": {
        "version": "2.0",
        "resultCode": "0",
```

```
        "resultMsg": "success"
    },
    "resultData": {
        "responseId": "9239cb50a8384bb0ab503441bf089446"
    },
    "resultCode": "0",
    "resultDesc": "success"
}
```


- Example of the message returned by the call center to the callback URL

  Message header:

```
POST / HTTP/1.1
Accept: */*
Content-Type: application/json;charset=UTF-8
Authorization: auth-v2/ak/2018-07-02T02:43:03Z/content-length;content-type;host
/c5bd683******************ef5747bd572c28
Host: 10.57.118.171
Content-Length: 208
```

  Message parameters:


```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "responseId": "9239cb50a8384bb0ab503441bf089446",
        "agentOprInfoFileName": "20180702_9239cb50a8384bb0ab503441bf089446.zip"
    }
}
```


4.1.7 Downloading Indexes of Agent Operation Details

Description

This interface is invoked to download the .zip package of agent operation detail record indexes (*yyyymmdd_ responseId*). You can view the agent operation details (*yyyymmddhhmmssSSS+Three-digit random number*_**agentOprInfo.csv**) corresponding to each .zip package.

Usage Description

- Prerequisites
  - Files have been generated at the backend by invoking the interface for generating indexes of agent operation details.

- You have passed the authorization and obtained the access URL.

- Usage restrictions
  Developers can only download files under their own accounts. The developer account (**appId**) must correspond to the data record file name (**agentOprInfoFileName**).
  The developer account (**appId**) is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/downloadAgentOprInfoFile

Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Mandatory	Value Type	Default Value	Description
1	version	Yes	String (1–32)	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameter in the request body

No.	Name	Mandatory	Value Type	Default Value	Description
1	agentOprInfoFileName	Yes	String (1–49)	None	Index file name of agent operation details. The name is obtained by invoking the interface for generating indexes of agent operation details. The value is in *yyyyMMdd_{responseId}*.**zip** format.

Table 3 msgBody parameter in the request body

No.	Name	Mandatory	Value Type	Default Value	Description
					For example, if the interface for generating indexes of agent operation details has been invoked on **2018-07-02**, and if the obtained recording index **responseId** is **9239cb50a8384bb0ab503441bf08944**, the value of this parameter is **20180702_9239cb50a8384bb0ab503441bf089446.zip**.

Response

If the interface is successfully invoked, the system obtains the binary data of the file from the HTTP response message to generate a CDR file. The file name extension is .zip. For details about the parameters in the file, see Definitions of CDR and Recording Index Files . In the response message, the value of **content-type** is **Application/Octet-stream;charset=UTF-8**.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see Table 4. **resultData** is a reserved field and is left empty by default.

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.
4	response	Object	Request result object. For details, see Table 5. This field is reserved for compatibility with earlier versions and is not recommended.

Table 5 Parameters in a response message

No.	Name	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, 2.0.
4.2	resultCode	String (1–32)	Result code returned. For details, see Error Code Reference .
4.3	resultMsg	String	Request result description.

Example

- Request header

```
POST /CCFS/resource/ccfs/downloadAgentOprInfoFile HTTP/1.1
```

```
Authorization: auth-v2/ak/2018-07-02T02:43:08Z/content-length;content-type;host
/ae066c2f5de*****************afc161cb7e66f5d
Accept: */*
Host: 10.154.198.164
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "agentOprInfoFileName": "20180702_9239cb50a8384bb0ab503441bf089446.zip"
    }
}
```

4.1.8 Generating Index Files of Agent Skill Change Details

Description

This interface is invoked to generate an index file of agent skill change details based on conditions. After a file is generated, the file name (**agentOprSkillFileName**) is sent to the callback URL carried by **callBackURL** in callback mode.

The callback of the URL specified by **callBackURL** requires authentication. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Usage Description

- Prerequisites
  - The developer environment has been established and calls can be made properly. The developer has an independent server and the server has been bound to the callback URL (**callBackURL**).
  - **accountId**, **agentId**, and **callId** are optional parameters, which can be flexibly combined to generate CDRs and recording index files. If none of the three parameters is specified, the corresponding data records of the developer in the specified period are generated.
  - You have passed the authorization and obtained the access URL.
  - You have set **aicc.ssl.trustAll** in the CC-FS configuration file **/home/ccfsapp/webapps/ccfsapp/WEB-INF/classes/config/servicecloud.base.properties** to **false** and performed the following steps to place the certificate:

    1. Log in to the CC-FS server as an O&M user and switch to the **root** user.

    ```
    su - root
    ```

2. Upload the third-party certificate that needs to be loaded by the callback URL for notifying the index file generated by the CC-FS to the same directory (**$HOME/conf**) as the **truststore.jks** certificate file of the AICC. The directory varies according to the actual environment.
3. Change the owner group of the uploaded third-party certificate.

```
chown ccfsapp:ccfsapp XXX.jks
```

4. Import the third-party certificate to *truststore.jks*.

```
keytool -import -alias xxx -file XXX.jks -keystore truststore.jks
```

   Enter the password of the *truststore.jks* certificate file as prompted.
5. Delete the *XXX.jks* certificate from the environment.
6. Restart the CC-FS.
   Go to the **/home/ccfsapp/bin** directory and run the following command:

```
./shutdown.sh;./startup.sh
```

- Usage restrictions
  A developer can access only the data of the account. The developer account (**appId**) must correspond to the enterprise account (tenant space ID: **accountId**).
  The developer account (**appId**) is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://*ip*:*port*/CCFS/resource/ccfs/queryAgentSkillInfoData
Set *IP address* to the IP address of the server where the CC-FS is installed and *Port number* to the HTTPS port number of the CC-FS.
If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request Description

Table 1 Parameters in the request message header

No.	Parameter	Value Type	Mandatory (Yes/No)	Default Value	Description
1	Content-Type	String	True	None	The value is fixed at **application/json; charset=UTF-8**.

Table 1 Parameters in the request message header

No.	Parameter	Value Type	Mandatory (Yes/No)	Default Value	Description
2	Authorization	String	True	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameters in the request body

No.	Parameter	Value Type	Mandatory (Yes/No)	Default Value	Description
1	version	String (1–32)	True	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameters in the request body

No.	Parameter	Value Type	Mandatory (Yes/No)	Default Value	Description
1	beginTime	String	True	None	Start time of an agent skill change, which is the time in the time zone where the tenant space is located. The interval between the start time and end time cannot exceed three days. The time is in *yyyy-MM-dd HH:mm:ss* format.
2	endTime	String	True	None	End time of an agent skill change, which is the time in the time zone where the tenant space is located. The interval between the start time and end time cannot exceed three days. The time is in *yyyy-MM-dd HH:mm:ss* format.
3	accountId	String	False	None	Enterprise account (tenant space ID).
4	agentId	String	False	None	Agent ID.
5	callBackUrl	String	True	None	Callback URL. After generating a data record file, the platform returns **fileName** to the URL specified by this parameter.

Response description:

Types:

- Response received immediately after this interface is invoked

For details about the parameters in the response message, see Table 4 .

Table 4 Parameters in the response message

No.	Parameter	Value Type	Description
1	resultCode	String	Result code returned. **0**: success Others: See Error Code Reference .

Table 4 Parameters in the response message

No.	Parameter	Value Type	Description
2	resultDesc	String	Request result description.
3	resultData	Object	Response data. For details about the **resultData** parameter, see Table 5 .
4	response	Object	Request result object. For details, see Table 6 . This field is reserved for compatibility with earlier versions and is not recommended.

Table 5 resultData parameter in the response

No.	Parameter	Value Type	Description
3.1	responseId	String	Unique ID generated after the interface is successfully invoked. It is a part of the file name.

Table 6 Parameters in the response message

No.	Parameter	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, **2.0**.
4.2	resultCode	String(1-32)	Result code returned. For details, see Error Code Reference .
4.3	resultMsg	String	Request result description.

- Response to third-party callback after data record file generation

The response consists of **request** and **msgBody**. For details about the **msgBody** parameter, see Table 7 .

request contains **version** (**2.0** by default).

msgBody consists of **responseId** (part of the agent operation CDR file name), **fileName** (name of the ZIP package of agent operation CDRs to be downloaded), and **accountId** (tenant space ID).

Table 7 msgBody parameters in the callback response body

No.	Parameter	Value Type	Description
1	responseId	String	Unique ID generated after the interface is successfully invoked. It is a part of the file name.
2	fileName	String	Name of a CDR or recording index file. The value is in *yyyyMMdd_{responseId}*.**zip** format. For example, if the date when this interface is invoked is 2018-07-02 and the obtained value of **responseId** is **9239cb50a8384bb0ab503441bf089446**, the value of this parameter is **20180702_9239cb50a8384bb0ab503441bf089446.zip**.
3	accountId	String	Enterprise account (tenant space ID). The enterprise account carried in the request parameter is used to associate the file name with an enterprise. If **accountId** is not set in the request parameter, this value is not returned.

NOTE

If the callback fails (the CC-FS does not receive any success response from the callback URL or the response body is missing), the CC-FS retries at an interval of 10 minutes until the callback is successful or the number of retry times exceeds the upper limit.

Example

- Request header

```
POST /CCFS/resource/ccfs/ivr/deleteFile  HTTP/1.1
Authorization: auth-v2/ak/2021-08-31T09:38:50.872Z/content-length;content-
type;host/c12f0ed0******************941bdd106
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "beginTime": "2018-06-29 10:42:49",
        "endTime": "2018-07-02 10:42:49",
        "accountId": "",
        "agentId": "",
        "callBackURL": "https://10.57.118.171:8000"
    }
}
```

- Response parameters

```
{
    "resultData": null,
    "resultCode": "0",
    "resultDesc": "success"
}
```

4.1.9 Downloading Index Files of Agent Skill Change Details

Description

Download the ZIP packages (*yyyymmdd_responseId*) of index files of agent skill change details to view the file (*yyyymmddhhmmssSSS* + *three-digit random number*_**agentSkillInfo.csv**) of agent skill change details in each ZIP package.

Usage Description

- Prerequisites
  - Files have been generated at the backend by invoking the interface for generating indexes of agent skill change details.
  - You have passed the authorization and obtained the access URL.
- Usage restrictions
  Developers can only download files under their own accounts. The developer account (**appId**) must correspond to the data record file name (**agentOprInfoFileName**).
  The developer account (**appId**) is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://*ip*:*port*/CCFS/resource/ccfs/downloadAgentSkillInfoFile
Set *IP address* to the IP address of the server where the CC-FS is installed and *Port number* to the HTTPS port number of the CC-FS.
If the request is routed through the NSLB, set *IP address* to the IP address of the NSLB server and *Port number* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory (Yes/No)	Default Value	Description
1	Content-Type	String	True	None	The value is fixed at **application/json; charset=UTF-8**.
2	Authorization	String	True	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Request parameter description

No.	Parameter	Value Type	Mandatory (Yes/No)	Default Value	Description
1	agentSkillInfoFileName	String	True	None	Name of an index file of agent skill change details It is obtained by invoking the **Agent Skill Details Index Files** API. The value is in *yyyyMMdd_{responseId}*.**zip** format. Example: The date when this interface is invoked is 2023-04-06. The obtained value of **responseId** is **9239cb50a8384bb0ab503441bf089446**. The value of this parameter is **20230406_9239cb50a8384bb0ab503441bf089446.zip**.

Response Description

If the interface is successfully invoked, the system obtains the binary data of the file from the HTTP response message to generate a CDR file whose file name extension is **.zip**. For details about the parameters in the CDR file, see the definitions in the index file of agent skill change. In the response, the **content-type** field is **Application/Octet-stream;charset=UTF-8**.

If the interface fails to be invoked, an error code is returned.

Table 3 Parameters in the response message

No.	Parameter	Value Type	Description
1	resultCode	String	Result code returned. **0**: success Others: See Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.

Example

- Request header

```
POST /CCFS/resource/ccfs/ivr/deleteFile  HTTP/1.1
Authorization: auth-v2/ak/2021-08-31T09:38:50.872Z/content-length;content-type;host/c12f0ed0******************941bdd106
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "agentSkillInfoFileName": "20180702_9239cb50a8384bb0ab503441bf089446.zip"
    }
}
```

- Response parameters

```
{
    "resultData": null,
    "resultCode": "0",
    "resultDesc": "success"
}
```

4.1.10 Requesting to Play Back a Recording File

Description

After this API is invoked, the backend verifies the recording file name. If the verification is successful, a token is returned for Playing Back a Recording File .

Usage Description

- Prerequisites
  - Recording index files have been obtained by invoking the interface for downloading CDRs and recording indexes.
  - You have passed the authorization and obtained the access URL.
- Usage restrictions
  Developers can only play recording files under their own accounts. The developer account (**appId**) must correspond to the recording file name (**fileName**).
  The developer account (**appId**) is contained in the authentication string. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/requestPlayVoice

Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	fileName	String	Yes	None	Relative path of the recording file, which is obtained from the **original_file_name** field in the recording index file returned when the interface for downloading CDRs and recording indexes is invoked. Note: The backslash (\) in **fileName** needs to be changed to a slash (/).

Response

If the interface is invoked successfully, a response is returned. In the response, the value of **resultCode** is **0**, and the value of **resultMsg** is **success**.

For details about the response data structure, see Table 4. The **resultData** parameter carries the token.

If the interface fails to be invoked, an error code is returned. The default value of **resultData** is **null**.

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned.

Table 4 Parameters in the response

No.	Name	Value Type	Description
			0: success Other values: For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data. For details, see Table 5.
4	response	Object	Request result object. For details, see Table 6. This field is reserved for compatibility with earlier versions and is not recommended.

Table 5 resultData parameter in the response

No.	Name	Value Type	Description
3.1	token	String	After this interface is successfully invoked, the token is used for authentication for Playing Back a Recording File . The token is valid for 30 minutes.

Table 6 response parameters in the response

No.	Name	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, 2.0.
4.2	resultCode	String (1–32)	Result code returned. For details, see Error Code Reference .
4.3	resultMsg	String	Request result description.

Example

- Request header

```
POST /CCFS/resource/ccfs/requestPlayVoice HTTP/1.1
Authorization: auth-v2/taikang/2018-07-02T02:42:49Z/content-length;content-type;host
/7a8fb9d6******************5df0cce3c6a253
Accept: */*
Host: 10.154.198.164:18084
Content-Type: application/json;charset=UTF-8
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
```

```
    "msgBody": {
        "fileName": "Y:/1/0/20180309/108/2022318.V3"
    }
}
```

- Response header

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
Date: Mon, 02 Jul 2018 02:43:03 GMT
```

- Response parameters


```
{
    "response": {
        "version": "2.0",
        "resultCode": "0",
        "resultMsg": "success"
    },
    "resultData": {
        "token": "DC45F*****************6D46F90"
    },
    "resultCode": "0",
    "resultDesc": "success"
}
```


4.1.11 Playing Back a Recording File

Description

This interface is invoked by the Windows Media Player to play recording files in URL mode.

Usage Description

- Prerequisites
  - You have obtained the token by invoking the interface described in Requesting to Play Back a Recording File .
  - The access URL has been obtained.
- Usage restrictions
  Developers can only download files under their own accounts. The developer account (**appId**) must correspond to the recording file name (**fileName**).

Interface Method

GET

URI

https://ip:port/CCFS/resource/ccfs/playVoice

Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request

No.	Name	Value Type	Mandatory	Default Value	Description
1	t	String	Yes	None	To prevent data cache, you are advised to use the current timestamp.
2	version	String	Yes	2.0	Protocol version. The default value is **2.0**.
3	st	String	Yes	None	Token returned when the interface described in Requesting to Play Back a Recording File is invoked.

Response

If this interface is successfully invoked, the binary data stream of the corresponding recording file is returned, which can be directly played using the Windows media player.

If the interface fails to be invoked, an error code is returned. In the response, the value of **content-type** is **application/octet-stream;charset=UTF-8**. For details about the error code response data structure, see Table 2. **resultData** is a reserved field and is defaulted to **null**.

Table 2 Parameters in a response message

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.
4	response	Object	Request result object. For details, see Table 3. This field is reserved for compatibility with earlier versions and is not recommended.

Table 3 Parameters in a response message

No.	Name	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, 2.0.
4.2	resultCode	String (1–32)	Result code returned. For details, see Error Code Reference .
4.3	resultMsg	String	Request result description.

Example

- Request

```
GET
https://ip:port/CCFS/resource/ccfs/playVoice?t=1596677685425&version=2.0&st=7D41CF5DE98948979AAE063114C8AFCF3
```

- Response header

```
HTTP/1.1 200 OK
Content-Type: application/octet-stream;charset=UTF-8
Date: Mon, 02 Jul 2018 02:43:03 GMT
```

- Response parameters

```
{
    "resultData": null,
    "resultCode": "010106",
    "resultDesc": "Parameter token is empty."
}
```

4.1.12 Downloading an Intelligent IVR Recording File

Description

This interface is invoked to download an intelligent IVR recording file.

Usage Description

- Prerequisites
  - You have passed the authorization and obtained the access URL.
- Usage restrictions
  Developers can download only files under their own accounts. The VDN ID corresponding to the developer account (**appId**) must contain the VDN ID in the intelligent IVR recording file name (**fileName**).

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/downloadOiapRecord?locationId={locationId}

Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Table 1 Parameters in the URL

No.	Name	Value Type	Mandatory	Description
1	locationId	Integer	No	Node ID. The value can be obtained using the session record query interface https://*IP*.*PORT*/oifde/rest/api/queryRecordHistory.

Request

Table 2 Parameters in the request

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 3 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 4 msgBody parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	fileName	String (1–127)	Yes	None	Name of an intelligent IVR recording file. Parameter format: 1: *{Recording drive letter}*:/*{record}*/*{VDN ID}*/*{odfsrecord}*/*{yyyymmdd}*/*{Main file name}*.wav Example: **Y:/record/39/odfsrecord/20201015/10233946082696733.wav**

Table 4 msgBody parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
					2: *{Recording drive letter}*:/*{VDN ID}*/*{odfsrecord}*/*{yyyymmdd}*/*{Main file name}*.wav Example: **Y:/39/odfsrecord/20201015/10233946082696733.wav** 3: *{Recording drive letter}*:/*{record}*/*{VDN ID}*/*{yyyymmdd}*/*{Main file name}*.wav Example: **Y:/record/39/20201015/10233946082696733.wav** 4: *{Recording drive letter}*:/*{VDN ID}*/*{transferrecord}*/*{yyyymmdd}*/*{Main file name}*.wav Example: **Y:/39/transferrecord/20201015/10233946082696733.wav** Note: *{VDN ID}* indicates the VDN ID, *{yyyymmdd}* indicates the date when the intelligent IVR recording file is generated, and *{Main file name}* indicates the name of the recording file.

Response

If the interface is successfully invoked, the system obtains the binary data of the file from the HTTP response message to generate a recording file. The file name extension is .wav. In the response message, the value of **content-type** is **Application/Octet-stream;charset=UTF-8**.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see Table 5. **resultData** is a reserved field and is left empty by default.

Table 5 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.
4	response	Object	Request result object. For details, see Table 6. This field is reserved for compatibility with earlier versions and is not recommended.

Table 6 Parameters in a response message

No.	Name	Value Type	Description
4.1	version	String (1–32)	Protocol version, for example, 2.0.
4.2	resultCode	String (1–32)	Result code returned. For details, see Error Code Reference .
4.3	resultMsg	String	Request result description.

Example

- Request header

```
POST /CCFS/resource/ccfs/downloadOiapRecordHTTP/1.1
Authorization: auth-v2/ak/2018-07-02T02:45:50Z/content-length;content-type;host/
eb453f68e******************96c509c4913
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "fileName": "Y:/39/odfsrecord/20201019/1603089033-1205143033.wav"
    }
}
```

4.1.13 Downloading IVR Voice Messages

Description

This interface is invoked to download a recording file of IVR voice messages.

Usage Description

- Prerequisites
  - You have passed the authorization and obtained the access URL.
- Usage restrictions
  Developers can download only files under their own accounts. The VDN ID corresponding to the developer account (**appId**) must contain the VDN ID in the recording file name (**fileName**) of IVR voice messages.

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/downloadIVRRecordFile?locationId={locationId}

Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Table 1 Parameters in the URL

No.	Name	Value Type	Mandatory	Description
1	locationId	Integer	No	Node ID. The value can be obtained using the session record query interface https://*IP.PORT*/oifde/rest/api/queryRecordHistory.

Request

Table 2 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 3 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 4 msgBody parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	fileName	String (1–225)	Yes	None	Recording file name of IVR voice messages. Parameter format: *{Recording drive letter}*:/*{VDN ID}*/**record**/*{yyyymmdd}*/*{Main file name}*.**wav** Example: **Y:/39/record/20201015/10233946082696733.wav** Note: *{VDN ID}* indicates the VDN ID, *{yyyymmdd}* indicates the date when the recording file of IVR voice messages is generated, and *{Main file name}* indicates the name of the recording file. The directory after *{VDN ID}* must be the value of the system parameter **Directory next to vdnId in IVR message recording files**. Contact the system administrator to obtain the value of the system parameter in **Configuration Center** > **System Management** > **System Parameter Configuration** > **System parameters** > **File storage service** > **Path Configuration** > **Directory next to vdnId in IVR message recording files**. The default value is **record**.

Response

If the interface is successfully invoked, the system obtains the binary data of the file from the HTTP response message to generate a recording file. The file name extension is .wav. In the response message, the value of **content-type** is **Application/Octet-stream;charset=UTF-8**.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see the following table. **resultData** is a reserved field and is left empty by default.

Table 5 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.

Example

- Request header

```
POST /CCFS/resource/ccfs/downloadRecordFile HTTP/1.1
Authorization: auth-v2/ak/2018-07-02T02:45:50Z/content-length;content-type;host/
eb453f68e858*****************196c509c4913
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "fileName": "Y:/39/record/20201015/10233946082696733.wav"
    }
}
```

4.1.14 Downloading a Recording File as a Third Party

Description

This interface is invoked by a third-party to download a recording file. The third-party system is connected to the IVR system by loading a customized JAR package.

Usage Description

- Prerequisites
  - You have passed the authorization.
- Usage restrictions
  Developers can download only files whose path is the value of the system parameter **Third-Party Recording Download Path Rule** under their own accounts. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication . Contact the system administrator to obtain the value of the system parameter in **Configuration Center** > **System Management** > **System Parameter Configuration** > **System parameters** > **File storage service** > **Path Configuration** > **Third-Party Recording Download Path Rule**.

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/ivr/downloadFile
Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.
If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameters in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	filePath	String	Yes	None	If **mode** is set to **1**, **filePath** indicates the path of the folder where the file is located, for example, **Y:/voice/1/IVRrecordFile**. If **mode** is not specified or is set to another value, the value of **filePath** must be a full file path, for example, **Y:/voice/1/IVRrecordFile/xx.wav**. Note: The file path is specified by the system parameter **Third-Party Recording Download Path Rule**. Contact the system administrator to obtain the value of the system parameter in **Configuration Center** > **System Management** > **System Parameter Configuration** > **System parameters** > **File storage service** > **Path Configuration** > **Third-Party Recording Download Path Rule**. Default value: **/voice/ %VDNNO %/IVRrecordFile;/3rdfile/ %VDNNO %/IVRrecordFile;/3rdfile/ %VDNNO %/thirdvoicebotfile**.
2	mode	Integer	No	None	If **mode** is set to **1**, the value of **filePath** is a directory that contains a unique file. If the directory does not contain any file or contains multiple files, a failure message is returned. If **mode** is not specified or is set to another value, the value of **filePath** must be a full file path containing the file name.

Response

If the interface is successfully invoked, the system obtains the binary data of the file from the HTTP response message to generate a recording file. The file name extension is .wav. In the response message, the value of **content-type** is **Application/Octet-stream;charset=UTF-8**.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see Table 4. **resultData** is a reserved field and is left empty by default.

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.

Example

- Request header

```
POST /CCFS/resource/ccfs/ivr/downloadFile HTTP/1.1
Authorization: auth-v2/ak/2021-08-31T09:38:50.872Z/content-length;content-
type;host/c12f0ed0****************94941bdd106
```

```
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters


```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "filePath": "Y:/3rdfile/53/thirdvoicebotfile",
        "mode":"1"
    }
}
```


4.1.15 Uploading a File as a Third Party

Description

This interface is invoked by a third party to upload a file for a third-party chatbot. The third-party system is connected to the IVR system by loading a customized JAR package. The SaaS and OP deployment modes are supported.

Usage Description

- Prerequisites
  - You have passed the authorization.
- Usage restrictions

  Developers can upload files to only folders under their own accounts. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

  Contact the system administrator to obtain the upload path in **Configuration Center** > **System Management** > **System Parameter Configuration** > **System parameters** > **File storage service** > **Path Configuration** > **Third-Party Recording Download Path Rule**.

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/ivr/uploadFile?vdnId={vdnId}

Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.

If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Table 1 Parameters in the URL

No.	Name	Value Type	Mandatory	Description
1	vdnId	Integer	Yes	ID of a VDN.

Request

Table 2 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 3 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 4 msgBody parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	data	String	Yes	None	Base64 code of the file to be uploaded. Note: Only WAV voice files in the following format are supported: 8 bit, 8000 Hz, 64 kbit/s, and mono. The file size is specified by the system parameter **Third-Party Upload File Size Limit**, in MB. The default value is **5**. To modify the file size, contact the system administrator to set the system parameter in **Configuration Center** > **System Management** > **System Parameter Configuration** > **System parameters** > **File storage service** > **Resource allocation** > **Third-Party Upload File Size Limit**.

Response

If this interface is successfully invoked, **0** and the message "operate success" are returned.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see Table 4 . **resultData** is a reserved field and is left empty by default.

Table 5 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data. For details, see Table 6.

Table 6 Description of resultData in the response message

No.	Name	Value Type	Description
3.1	fileName	String	Name of the file to be uploaded. The file name must be the value of the system parameter **Third-Party Upload File Path Rule**. Contact the system administrator to obtain the value of the system parameter in **Configuration Center** > **System Management** > **System Parameter Configuration** > **System parameters** > **File storage service** > **Path Configuration** > **Third-Party Upload File Path Rule**. The default value is **/3rdfile/%VDNNO%/thirdvoicebotfile/**. Example: *{Drive letter}*:**/3rdfile/***{VDN ID}***/thirdvoicebotfile/***xxx***.wav**
3.2	locationId	List <String>	In CTI pool mode, the information is displayed if the upload is successful:

Example

- Request header

```
POST /CCFS/resource/ccfs/ivr/upload?vdnId=xx HTTP/1.1
Authorization: auth-v2/ak/2021-08-31T09:38:50.872Z/content-length;content-
type;host/c12f0ed*****************941bdd106
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "data":"base64"
    }
}
```

- Response parameters

```
{
```

```
    "resultData": {
        "fileName": "Y:/3rdfile/53/thirdvoicebotfile/8fSsBaEb_1631869974029.wav",
        "locationId": [
            "0",
            "256"
        ]
    },
    "resultCode": "0",
    "resultDesc": "success"
}
```


4.1.16 Generating a File Hash Value as a Third Party

Description

This API is invoked by a third party to generate the hash value of an uploaded file. The third-party system is connected to the IVR system by loading a customized JAR package.

Usage Description

- Prerequisites
  - You have passed the authorization.
- Usage restrictions
  Developers can obtain only the hash values of the files under their own accounts. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/ivr/getFileHash
Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.
If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameters in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	fileName	String	Yes	None	Full path of the file whose hash value needs to be obtained. The file path is specified by the system parameter **Third-Party Upload File Path Rule**. Contact the system administrator to obtain the value of the system parameter in **Configuration Center** > **System Management** > **System Parameter Configuration** > **System parameters** > **File storage service** > **Path Configuration** > **Third-Party Upload File Path Rule**. The default value is **/3rdfile/%VDNNO%/thirdvoicebotfile/**. Example: *{Drive letter}*:**/3rdfile/***{VDN ID}***/thirdvoicebotfile/***xxx***.wav**
2	locationId	String	No	None	This parameter is valid only in CTI pool mode.

Response

If this interface is successfully invoked, **0** and the message "operate success" are returned.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see Table 4 . **resultData** is a reserved field and is left empty by default.

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data. For details, see Table 5.

Table 5 Description of resultData in the response message

No.	Name	Value Type	Description
3.1	hashValue	String	String of the voice file after SHA256 encoding.

Example

- Request header

```
POST /CCFS/resource/ccfs/ivr/getFileHash HTTP/1.1
Authorization: auth-v2/ak/2021-08-31T09:38:50.872Z/content-length;content-
type;host/c12f0ed0******************494941bdd106
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "fileName": "Y:/3rdfile/2/thirdvoicebotfile/5dckNDei_1647502396161.wav"
    }
}
```

- Response parameters

```
{
    "resultData": {
        "hashValue": "2b67748fe335617******************cf19f28bb8"
    },
    "resultCode": "0",
    "resultDesc": "success"
}
```

4.1.17 Deleting an Uploaded File as a Third Party

Description

This API is invoked by a third party to delete an uploaded file. The third-party system is connected to the IVR system by loading a customized JAR package.

Usage Description

- Prerequisites
  - You have passed the authorization.
- Usage restrictions
  Developers can delete only files under their own accounts. For details, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Interface Method

POST

URI

https://ip:port/CCFS/resource/ccfs/ivr/deleteFile
Set *ip* to the IP address of the server where the CC-FS is installed and *port* to the HTTPS port number of the CC-FS.
If the request is routed through the NSLB, set *ip* to the IP address of the NSLB server and *port* to the HTTPS port number of the CC-FS service mapped on the NSLB.

Request

Table 1 Parameters in the request header

No.	Name	Value Type	Mandatory	Default Value	Description
1	Content-Type	String	Yes	None	The value is fixed to **application/json; charset=UTF-8**.
2	Authorization	String	Yes	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 request parameter in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	version	String (1–32)	Yes	2.0	Protocol version. Currently, the value is fixed to **2.0**.

Table 3 msgBody parameters in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
1	fileName	String	Yes	None	File to be deleted. The file path is specified by the system parameter **Third-Party Upload File Path Rule**. Contact the system administrator to obtain the value of the system parameter in **Configuration Center** > **System Management** > **System Parameter Configuration** > **System parameters** > **File storage service** > **Path Configuration** > **Third-Party Upload File**

Table 3 msgBody parameters in the request body

No.	Name	Value Type	Mandatory	Default Value	Description
					Path Rule. The default value is **/3rdfile/%VDNNO%/thirdvoicebotfile/**. Example: *{Drive letter}*:**/3rdfile/***{VDN ID}***/thirdvoicebotfile/***xxx***.wav**
2	locationId	String	No	None	This parameter is valid only in CTI pool mode. If **locationId** has a value, delete the files in the path specified by **uapId** corresponding to **locationId**. If **locationId** is empty, delete all files.

Response

If this interface is successfully invoked, **0** and the message "operate success" are returned.

If the interface fails to be invoked, an error code is returned. For details about the error code data structure, see Table 4 . **resultData** is a reserved field and is left empty by default.

Table 4 Parameters in the response

No.	Name	Value Type	Description
1	resultCode	String	Result code returned. For details, see Error Code Reference .
2	resultDesc	String	Request result description.
3	resultData	Object	Response data.

Example

- Request header

```
POST /CCFS/resource/ccfs/ivr/deleteFile  HTTP/1.1
Authorization: auth-v2/ak/2021-08-31T09:38:50.872Z/content-length;content-type;host/c12f0ed0****************941bdd106
Accept: */*
Content-Type: application/json;charset=UTF-8
Content-Length: 193
```

- Request parameters

```
{
    "request": {
        "version": "2.0"
    },
    "msgBody": {
        "fileName": "Y:/3rdfile/2/thirdvoicebotfile/5dckNDei_1647502396161.wav"
```

```
    }
}
```

- Response parameters

```
{
    "resultData": null,
    "resultCode": "0",
    "resultDesc": "success"
}
```

4.2 Index File Definition

4.2.1 Definitions of CDR and Recording Index Files

Definitions of Recording Index Files (*yyyymmddhhmmssSSS*+*3-digit random number*_record_*file sequence number*.csv)

In the exported .csv recording file, each line indicates a piece of data. The data in each line is sorted by column. The following table describes the data in each line.

No.	Parameter	Type	Description
1	developer	String (1–64)	ID of a developer.
2	account_id	String (1–64)	Enterprise account.
3	cc_id	String (1–64)	ID of a call center.
4	agent_id	String (1–32)	Agent ID.
5	call_id	String (1–25)	Call ID.
6	caller_no	String (1–25)	Calling number.
7	callee_no	String (1–25)	Called number.
8	call_begin	Date	Start time. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.
9	call_end	Date	End time. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.

No.	Parameter	Type	Description
10	file_name	String (1–255)	Name of a recording file, for example, **/10/1/record/100/13533/0903379.wav**.
11	task_status	String (1–20)	Indicates whether to convert a recording file to a text file. The options are **1** (yes) and **0** (no).
12	task_result	String (1–20)	Whether a recording file is successfully converted into a text file. The options are **success** and **failed**.
13	original_file_name	String (1–129)	Path to the original recording file. Example: **X:\17\0\20180903\15470\2043295.V3**
14	vdn_id	String (1–64)	ID of the VCC that processes a call.
15	call_type	String (1–64)	Call type. Possible types: Inbound call Outbound call For details, see Description of Call Types (Recording Index) .
16	media_type	String (1–64)	Media type. For details, see Media Type .
17	user_wanted_skill_id	String (1–64)	Direction skill.
18	current_skill_id	String (1–64)	ID of the skill queue that processes a call.

Definitions of CDR Files (*yyyymmddhhmmssSSS*+*3-digit random number*_call_*file sequence number*.csv)

In the exported .csv CDR file, each line indicates a piece of data. The data in each line is sorted by column. The following table describes the data in each line.

No.	Parameter	Type	Description
1	developer	String (1–64)	ID of a developer.
2	account_id	String (1–64)	Enterprise account.
3	cc_id	String (1–64)	ID of a call center.
4	agent_id	String (1–32)	Agent ID.
5	call_id	String (1–25)	Call ID.

No.	Parameter	Type	Description
6	caller_no	String (1–26)	Calling number.
7	callee_no	String (1–26)	Called number.
8	wait_begin	Date	Waiting start time. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.
9	wait_end	Date	Waiting end time. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.
10	ack_begin	Date	Response start time. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.
11	ack_end	Date	Response end time. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.
12	call_begin	Date	Call start time. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.
13	call_end	Date	Call end time. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.
14	device_type	Integer (1–4)	Device type. The options are **1** (skill queue), **2** (agent), and **3** (IVR flow).
15	device_no	Integer (1–5)	Device number. If the device type is skill queue, the device number is the skill queue ID. If the device type is agent, the device number is the agent ID. If the device type is IVR, the device ID is the IVR flow ID.
16	call_type	Integer (1–4)	Call type. For details, see Description of Call Types .
17	release_cause	Integer (1–5)	Call release cause. For details, see Cause Code . For the bill in which the value of **CallIDNUM** is **-1** (refer to item 18 in this table), this field indicates the call release reason. In other cases, this field is meaningless.
18	call_id_num	Integer (1–4)	Sequence number of a call ID. If there is only one call ID, that is, the call ID is unique, the sequence number is **-1**. If there are multiple records with the same call ID, the sequence number of the last record is **-1**, and those of other records are numbered in sequence. For example, if four records have the same call ID, and the values of **callidnum** are **1**, **2**, **3**, and **-1**.

No.	Parameter	Type	Description
19	vdn_id	String (1–32)	ID of the VCC that processes a call.
20	pre_device_type	String (1–32)	Type of the previous device that a call passes through. The options are **1** (skill queue), **2** (agent), and **3** (IVR).
21	pre_device_no	String (1–32)	ID of the previous device that a call passes through.
22	skill_id	String (1–32)	ID of the skill queue to which a call belongs. When a call is transferred because the skill queue to which the call belongs overflows or no agent is on duty, the value is the ID of the first skill queue that the call enters.
23	current_skill_id	String (1–32)	ID of the skill queue that processes a call.
24	device_in	String (1–32)	Description of the current device. For a skill queue, the value is **DEVICE_QUEUE**. For an agent, the value is **DEVICE_AGENT**. When the value of **DeviceType** is **IVR process** (3), the value is the IVR flow access code. For a call, the value is a phone number. For a temporary routing device, the value is **DEVICE_ROUTE**. For a virtual device, the value is **DEVICE_VNR**. When the value of **DeviceType** is **CDN device** (10), the value is the CDN number. When the value of **DeviceType** is **PGR** (14), the value is the PGR access code.
25	pre_device_in	String (1–32)	Description of the previous device. The values are as follows: For a skill queue, the value is **DEVICE_QUEUE**. For an agent, the value is **DEVICE_AGENT**. When the value of **DeviceType** is **IVR process** (3), the value is the IVR flow access code. For a call, the value is a phone number. For a temporary routing device, the value is **DEVICE_ROUTE**. For a virtual device, the value is **DEVICE_VNR**. When the value of **DeviceType** is **CDN device** (10), the value is the CDN number. When the value of **DeviceType** is **PGR** (14), the value is the PGR access code.
26	service_no	String (1–32)	Business type. Type of the business currently provided by a device. **NOTE:** This parameter has no actual service meaning. Do not use it.
27	leave_reason	String (1–32)	Reason why a call is disconnected from a device. For details, see Reasons for Leaving the Device of the Call .

No.	Parameter	Type	Description
28	media_type	String (1–32)	Media type of a call. For details, see Description of Call Media Types .
29	sub_media_type	String (1–32)	Submedia type. For details, see Table 1 . This parameter is valid only when **media_type** (call media type) is set to **MEDIA_TYPE_WEB_LEAVE_MESSAGE** (20).

4.2.2 Definition of Index Files for Agent Operation Details

Definitions of Agent Operation Bill Index Files (*yyyymmddhhmmssSSS*+*3-digit random number*_agentOprInfo_*file sequence number*.csv)

In the exported .csv file, each line indicates a piece of data. The data in each line is sorted by column. The following table describes the data in each line.

No.	Parameter	Type	Description
1	id	String (1–32)	Unique ID.
2	developer_id	String (1–64)	ID of a developer.
3	account_id	String (1–64)	Enterprise account.
4	agent_id	String (1–32)	Agent ID.
5	service_no	String (1–25)	Business type ID. **NOTE:** This parameter has no actual service meaning. Do not use it.
6	skill_id	String (1–25)	ID of a skill queue.
7	call_id	String (1–25)	Call ID.
8	begin_time	Date	Start time of an operation. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.
9	end_time	Date	End time of an operation. The time format is *YYYY-MM-DD hh:mm:ss*, for example, **2015-02-08 20:23:30**.
10	op_type	String (1–20)	Operation type. For details, see Table 1 .
11	op_object	String (1–20)	Operation object.

No.	Parameter	Type	Description
12	op_cause	String (1–20)	Operation reason.
13	media_type	String (1–20)	Media type.
14	vdn_id	String (1–20)	VCC ID.
15	cc_id	String (1–20)	ID of a call center.
16	agent_type	String (1–20)	Agent type or customer level.
17	phone_no	String (1–32)	Agent IP address, agent phone number, called number, or conference ID.
18	callIn_out	String (1–20)	Incoming or outgoing call flag. The options are as follows: Incoming Outgoing Outbound call made by IVR Outbound
19	location_id	String (1–20)	Distributed node ID.
20	logon_sn	String (1–25)	Sign-in SN.
21	skillInfo_sn	String (1–25)	Index of the skill list that an agent has when the agent performs the current operations. If the parameter value is blank, the agent does not have any skills. For example, the parameter value is blank when the agent logs in to the platform.

4.2.3 Definition of Agent Skill Details Index Files

Definition of Agent Skill Details Index Files (*yyyymmddhhmmssSSS* + *3-digit random number*_agentSkillInfo_*file sequence number*.csv)

In the exported .csv file, each line indicates a piece of data. The data in each line is sorted by column. The following table describes the data in each line.

No.	Parameter	Value Type	Description
1	id	String (1–32)	Unique ID.
2	developer_id	String (1–64)	Developer ID.
3	account_id	String (1–64)	Enterprise account.

No.	Parameter	Value Type	Description
4	sub_cc_no	Integer (1–11)	Call center ID.
5	vdn	Integer (1–11)	Sub call center ID.
6	agent_id	Integer (1–11)	Agent ID.
7	operate_time	DATE	Time of an operation in a tenant time zone. The time is in *YYYY-MM-DD hh:mm:ss* format, for example, **2015-02-08 20:23:30**.
8	operate_id	Integer (1–11)	Operator ID.
9	operate_type	Integer (1–11)	Operation type. For details, see Table 1 .
10	skill_list	String (1–2000)	Skill set, with IDs being separated by a comma (,).
11	logon_sn	String (1–26)	Sign-in sequence number.
12	operate_sn	Integer (1–11)	Operation sequence number, ranging from 1.
13	operate_end_time	DATE	End time of an operation in a tenant time zone. The time is in *YYYY-MM-DD hh:mm:ss* format, for example, **2015-02-08 20:23:30**.
14	renew_flag	Integer (1–11)	Update flag.
15	update_time	DATE	Time (GMT+00:00) when data is imported to the table.
16	month_of_year	Integer (1–4)	Month to which the time belongs.

5 Reference Description

5.1 Error Code Reference

Response Code	Result Code	Description	Handling Method
200	0	success	No handling is required.
200	0300001	The value of **{param}** cannot be null.	Check whether the parameter value in the request string is an empty string.
200	0300002	The value of **{param}** cannot be an empty string.	Check whether the parameter value in the request string is an empty string.

Response Code	Result Code	Description	Handling Method
200	0300003	The length of **{param}** is too long.	Check whether the parameter value is greater than the maximum length specified in the document.
200	0300004	The format of **{param}** is incorrect.	Check whether the parameter complies with the format described in the document.
200	0300005	The length of **{param}** is too short.	Check whether the parameter value is less than the maximum length specified in the document.
200	0300006	The value of version is error, current version is **{param}** .	Check whether the version number parameter is correct.
200	0300007	The params of requestBody is null.	Check whether the parameter length is matched.
200	0300008	Reached the download flow control threshold.	The maximum number of download requests that can be processed by the system is reached. Try again later.
200	0300009	Reached the query flow control threshold.	The maximum number of query requests that can be processed by the system is reached. Try again later.
200	0300010	The date in the value of fileName is in an incorrect format.	Check whether the file name in the request string is valid. The file name format is **/***{Node ID}***/***{Call center ID}***/record/***{App ID}***/***{Account ID}***/***{yyyymmdd}***/***{Agent ID}***/***{Original file name}***.***{File name extension}*. Check whether *{yyyymmdd}* is correct.
200	0300011	The file does not exist.	The requested file does not exist. Check whether the requested file is correct.
200	0300012	No data found.	Check whether the value of **fileName** is correct. The file name format is as follows: *{Drive letter}***:/***{VDN ID}***/***{***0** *(inbound call) or* **1** *(outbound call)}***/**{Date}**/***{Agent ID}***/***{Current time, in milliseconds}***.V3**
200	0300013	The beginTime cannot be later than the endTime.	Check the values of **beginTime** and **endTime** in the request.
200	0300014	The maximum duration between the beginTime and endTime is **{param}** days.	Check whether the time range specified by **beginTime** and **endTime** in the request string is valid. The maximum time range is 3 days by default.
200	0300015	The local storage or voice path does not exist.	The following configuration items in the configuration file must be properly set: **ccfs.common.local.store.path** **ccfs.common.local.voice.path**
200	0300016	The value of **{param}** is greater than current time.	Check the query interval.

Response Code	Result Code	Description	Handling Method
200	0300017	Specify ccId.	Multiple data records are found. The **ccId** parameter needs to be specified for more accurate query.
200	0300018	This record belongs to another developer.	The developer information in the recording to be queried does not match the developer information in the request. Check whether the value of **appId** is correct.
200	0300019	Specify at least one of fileName and callId.	Check whether **fileName** or **callId** is set.
200	0300020	Only one record is required but multiple records are obtained.	Multiple records are found based on the search criteria. However, only one record is required. Contact the administrator.
200	0300021	The **{param}** in fileName does not match the **{param}** in the directory.	Check whether the file name in the request string is valid. The file name format is */{Node ID}/{Call center ID}/***record/***{App ID}/{Account ID}/{yyyymmdd}/{Agent ID}/{Original file name}.{File name extension}*. Check whether *{App ID}*, **record**, and *{Node ID}* are correct.
200	0300022	The parameter **{param}** value contains illegal characters.	The value of the request parameter contains invalid special characters. Examples are as follows: The value of **billFileName** in the interface for downloading CDRs and recording indexes can contain only letters, digits, periods (.), hyphens (-), and underscores (_). The value of **fileName** in the interface for downloading recording files can contain only letters, digits, slashes (/), and periods (.). The value of **fileName** in the interface for downloading STT files can contain only letters, digits, slashes (/), and periods (.). The value of **fileName** in the interface for querying and downloading recording files can contain only letters, digits, colons (:), slashes (/), and periods (.)
200	0300023	The file name does not match the path configured in the configuration file.	During the download of IVR voice messages, the directory after the VDN ID in the path specified by the request parameter **fileName** does not match the value of the system parameter **Directory next to vdnId in IVR message recording files**. Set the system parameter to a proper file path.
200	0300024	The file path is not canonical path.	Check whether the path is a standard path.
200	0300025	Internal system error.	Contact the administrator.
200	0300026	Authentication failed.	Check whether the account ID or VDN ID is correct.
200	0300027	Parameter **{param}** is empty.	Check whether the parameter value is empty or an empty string.
200	0300028	Parameter **{param}** is error.	Check whether the parameter meets the requirements described in the document.

Response Code	Result Code	Description	Handling Method
200	0300029	File does not exist.	Check whether the value of **fileName** is correct. The file requested by the interfaces for requesting and playing back recording files does not exist.
200	0300030	Failure of system processing.	Contact the administrator.
200	0300031	Invalid oiap fileName, fileName Length is not legal.	Check the value length of **fileName**.
200	0300032	file path is invalid.	Check whether the file path is valid.
200	0300033	locationId is empty	Check whether the parameter value is empty or an empty string.
200	0300034	tenantId is empty	Check whether the parameter value is empty or an empty string.
200	0300035	audio file type should be wav.	Check the file format. Only audio files in WAV format are supported.
200	0300036	audio file channel should be mono.	Check the audio file channel.
200	0300037	audio file fileSampleSizeInBits should be 8000Hz.	Check the number of frames in a file.
200	0300038	audio file fileSampleRate should be 8000Hz.	Check the audio file sampling rate.
200	0300039	Failed to read the audio file format.	Check whether the uploaded audio file format meets requirements.
200	0300040	An exception occurred during file deletion.	Check whether the file permission is correct.
200	0300041	The file size exceeds the limit.	Check the file size.
200	0300042	file path config miss parameter %VDNNO%	Add the **VDNNO** variable to the path rule configuration item.
200	0300043	file path configuration should refer to the example.	Check the path content configuration item.
200	0300044	The directory contains multiple files.	Check whether the folder contains only one WAV file.
200	0300045	Failed to compress the recording files.	Check whether the recording files exist.

Response Code	Result Code	Description	Handling Method
200	0300046	The mount path corresponding to the drive letter does not exist.	Check the file path.
200	0300047	Upload file to OBS/LSS failed	Check whether the OBS/LSS service is normal.
200	0300048	In CTI pool mode,the value of ccfs.uap.id cannot be empty	Check whether the value of **ccfs.uap.id** in the **/home/ccfsapp/webapps/ccfsapp/WEB-INF/classes/config/servicecloud.base.properties** configuration file is empty.
200	0300049	In CTI pool mode,the value format of ccfs.uap.id is incorrect	Check whether the value format of **ccfs.uap.id** in the **/home/ccfsapp/webapps/ccfsapp/WEB-INF/classes/config/servicecloud.base.properties** configuration file is correct. The value format is *{Mount directory}*,*{UAP node ID}*.

5.2 Status Code Description

In the response, the status code is 200, indicating that the business is normal. Other status codes are returned as exception information.

Table 1 Status code description

Status Code	Description	Remarks
200	The business is normal.	For details, see the preceding interfaces.
401	Unauthorized access to the interface.	Check whether the authentication character string and request parameters are correct by referring to the interface authentication algorithm and Table 2. For any questions, contact the administrator.
403	The IP address is locked. The number of times that the authentication string carried in the request is incorrect exceeds the maximum number of allowed errors in a period.	Check whether the authentication character string and request parameters are correct by referring to the interface authentication algorithm and Table 2. For any questions, contact the administrator.

Table 2 Abnormal status code description

Parameter	Mandatory/Optional	Value Type	Remarks
timestamp	Mandatory	Long	Timestamp. Example: **1532142010247**.
status	Mandatory	int	Status code. Example: 401, 403
error	Mandatory	String	Error type. For example, **Forbidden** indicates that the permission is forbidden, and **Unauthorized** indicates that the permission is unauthorized.

Table 2 Abnormal status code description

Parameter	Mandatory/Optional	Value Type	Remarks
message	Mandatory	String	Prompt message. Example: **No message available**.
path	Mandatory	String	Requested path. Example: **/CCFS/resource/ccfs/queryBillData**

- Example of the returned error code


```
{"timestamp":1532142010247,"status":403,"error":"Forbidden","message":"Nomessageavailable","path":"/CCFS/resource/ccfs/queryBillData"}
```


5.3 Cause Code

Cause Code	Description
80	When a call is queuing in a skill queue, the call is transferred to the IVR flow for voice playback and digit collection. If a customer presses the cancel key, the flow uses this error code to request canceling the call queuing.
256	CCIVR/IVR normal call release.
263	The CCIVR/IVR disconnects a call because of connection failure.
338	The CCIVR/IVR disconnects a call because no flow can be found or an error occurs when parsing flows.
339	The CCIVR/IVR disconnects a call because there is no sufficient control blocks.
340	The CCIVR/IVR disconnects a call because the MRF script type is incorrect.
342	The CCIVR/IVR disconnects a call because the conference name is incorrect (possibly too long).
346	The CCIVR/IVR disconnects a call because waiting for the called party to pick up the phone times out after the CTIServer returns a call to the called party.
350	The CCIVR/IVR disconnects a call because waiting for the business module to pick up the phone times out.
354	The CCIVR/IVR disconnects a call because the handshake with the CTIServer times out.
356	The CCIVR/IVR disconnects a call because the CCIVR outgoing call times out.
357	The CCIVR/IVR disconnects a call because the IVR fails to connect the calling party and called party.
359	The CCIVR/IVR disconnects a call because the maximum conversation duration is exceeded.
360	The CCIVR/IVR disconnects a call because of other errors. For details, see the logs.
501	Default call release cause. The cause could be any of the following:

Cause Code	Description
	1. The routing result from the CCS is not received after a call accesses the platform. As a result, the call is disconnected. 2. After a call accesses the IVR, a flow disconnects the call.
502	The platform disconnects an outgoing call because waiting for the ringing of the ACD times out or waiting for call answer times out.
503	The platform disconnects a call because the handshake between the platform control block and ACD times out after the call is established between the ACD and the platform.
505	The cause could be any of the following: 1. When a call is transferred back to the agent console after the agent console transfers the call to the IVR in suspension transfer mode, the agent console times out in responding to the suspension resuming message. As a result, the platform disconnects the call. 2. When a call is transferred back to the IVR after the IVR transfers the call to the agent console in suspension transfer mode, the IVR times out in responding to the suspension resuming message. As a result, the platform disconnects the call.
506	After an agent transfers a call to the IVR in suspension transfer mode, waiting for the CCS to return the routing result times out.
507	After a non-agent device transfers a call to the IVR in suspension transfer mode, waiting for the CCS to return the routing result times out.
508	When a call is transferred, waiting for the CCS DO_FORWARD times out.
509	When an internal call is successfully transferred, waiting for the CCS FORWARD_COMIT times out.
510	When the platform initiates an outgoing call, waiting for the ACD SETUP_ACK message times out.
511	When the platform initiates an outgoing call, waiting for the CCS OCCUPY_ACK message times out.
512	A call is released because the number of calls waiting in a skill queue exceeds the maximum limit.
513	A call is released because the number of calls waiting in a skill queue exceeds the maximum limit.
514	When a call is transferred from an agent to an automatic flow, waiting for the automatic flow to respond times out.
515	During call transfer, waiting for the transfer destination device to respond times out.
516	Waiting for the routing response from the CCS times out.
517	Waiting for the ringing or response from the destination device times out.
518	The client disconnects a call.
519	When a call is transferred in suspension mode, the address sent by the CCS to DO_FORWARD is incorrect.
520	When a call is transferred in suspension mode, the address sent by the CCS to DO_FORWARD is incorrect.
521	An agent initiates call redirection (a call is disconnected and transferred out).

Cause Code	Description
522	An automatic flow initiates call redirection (a call is disconnected and transferred out).
523	When the platform initiates an outgoing call, the CCS returns a device occupation failure message.
525	The destination address in the routing result returned by the CCS is incorrect.
526	The CCS returns a routing failure message.
528	During outgoing call preview, the CCS returns a failure message.
529	After a flow transfers a call to an agent in suspension mode, if the agent hangs up and transfers the call back to the IVR, the IVR returns a failure message.
530	The IVR initiates call combination.
531	A call is normally disconnected in the ACD.
533	When a flow transfers a call to another flow, the flow releases the call before ringing.
534	When the CTIServer is disconnected from an ACD module, all the calls of the ACD module are released.
535	The CCS returns a routing failure message.
536	When a customer cancels queuing, the platform releases the call.
537	A call is disconnected in the ACD because the called party is busy.
538	A call is disconnected in the ACD because the called number does not exist.
539	A call is released after the CCS is switched.
540	A call is released after voice playing.
541	A call is transferred in suspension mode to a device. After the destination device processes the call, and the call is returned to the suspension device, the CCS returns the RESUME_ACK error message.
542	A call is transferred in suspension mode to a device. After the destination device processes and releases the call, the CCS returns the HUNG_RELEASE_ACK error message.
543	A call is transferred in suspension mode to a device. After the destination device processes the call, waiting for the CCS to send RESUME_ACK to notify the suspension device of call resumption times out.
544	A call is transferred in suspension mode to a device. After the destination device processes and releases the call, waiting for the CCS to return the release result HUNG_RELEASE_ACK times out.
547	When a call is transferred, the re-routing result returned by the CCS is not a virtual device, but the transfer indication address sent by the CCS is a virtual device.
548	A call is disconnected in the ACD because the mobile phone of the called party is powered off.
549	A call is disconnected in the ACD because the called party is out of the service area.

Cause Code	Description
550	When a network call is transferred in suspension mode, after the destination device processes and releases the call, the CCS returns an error.
551	A call is disconnected in the ACD for another reason.
552	During a network call, the peer call center is abnormal.
553	During a network call, the destination call center times out in waiting for the network incoming call from the ACD.
561	A call is released due to queue overflow.
562	No agent is on duty.
564	The number of concurrent calls exceeds the configured limit.
565	The total number of automatic calls exceeds the limit.
566	The number of concurrent calls in the flow exceeds the maximum.
567	The called number cannot be found.
568	The customer level is incorrect.
601	The ACD disconnects a call because the calling party hangs up before the call is answered.
602	The ACD disconnects a call because no resource is available.
603	The ACD disconnects a call because the called party does not answer the call.
604	The ACD disconnects a call because the called party rejects the call.
605	The ACD disconnects a call because the line is busy.
606	The ACD disconnects a call because the call times out.
607	The ACD disconnects a call because the mail box is full or suspended. For details, see the UAP original call release code 5.
608	The ACD disconnects a call because emails cannot be created for calls. For details, see the UAP original call release code 6.
609	The ACD disconnects a call because the channel is reset.
610	The ACD disconnects a call because of abnormal UAP resource operations.
611	The ACD disconnects a call because the Cp conference is faulty.
612	The ACD disconnects a call because the Intess control block is faulty.
613~619	The ACD disconnects a call.
714	Internal routing fails because IVR resources are occupied or no IVR access code is registered.

Cause Code	Description
732	The cause could be any of the following: 1. During a conference call, a participant is removed from the conference. 2. During a conference call, the conference moderator proactively releases the conference or the CCS is abnormal.
852	The client disconnects and rejects a call.
853	A call is released because an agent does not answer the call for a long time.
1057	Failed to route the message leaving call.
1058	Waiting for the routing result of the message leaving call times out.
1059	Waiting of a message leaving call for the queuing message times out.
1060	Waiting for an agent to answer a message leaving call times out.
1061	The release of a message leaving call times out.
1062	The internal message leaving call times out.
1063	The handshake of the message leaving call times out.
1064	Failed to create the conference.
1065	Conference creation times out.
1066	Failed to join the conference.
1067	Querying the enterprise ID times out.
1068	Failed to query the enterprise ID.
1069	The value of **EnterPriseID** is empty.
1070	Querying conference information times out.
1071	Failed to query conference information.
1072	Obtaining conference data times out.
1073	Failed to obtain conference data.
1074	Conference detection times out.
1075	The number of participants in the conference is less than expected.
1076	Releasing the conference times out.
1077	The call duration reaches the maximum.

5.4 Agent Operation Type

Table 1 Agent operation type

Operation Type	Description	Operation Object	Operation Reason
0	The agent signs out of CCS.	-1	1-255: The error is caused by the client. 256: The agent is disconnected from the platform. 257: The agent is forcibly signed out by the inspector. 258: The agent signs out. 259: The agent signs out before signs in again. 260: The handshake between the agent control blocks time out between the media server and the CCS. 261: The CCS detects that the handshake with the UIS about the agent control block times out. 262: The handshake between the media server and the CCS is disconnected. As a result, the agent signs out. 263: The handshake between the UIS and the agent times out. 264: An agent with the same employee ID is forcibly signing in to a new client. As a result, the agent is forcibly signed out.
1	An agent signs in.	-1	-1
2	Idle.	-1	-1
3	Busy.	-1	1: Indicates that an agent indicates busy. 2: An agent does not answer a call in a long period, so the agent is forced to show busy. 3: An agent is forced to show busy by a QC inspector. 4: Indicates that an agent takes a rest. 5: The phone of an agent is unavailable. 7: An agent rejects a call. 200-250: error codes customized by the service side.
4	Talking.	-1	1: The agent does not really exit the conversation status. Bills are generated every five minutes starting from 00:00 to 23:55. 0: The agent fully exits the conversation status. The conversation status in these bills refers to only text conversations.
10	Three-party conversation.	Employee ID.	-1
13	Manual station to automatic station (an agent be in the Suspended state).	-1	-1

Table 1 Agent operation type

Operation Type	Description	Operation Object	Operation Reason
20	Listening/Insert/Whisper	Employee ID.	0
22	Play voice.	-1	-1
26	Be on holiday or take a rest.	Employee ID.	The rest cause code is specified when the client invokes CCC_REQUEST_REST_EX_Msg. The default cause code is 0.
27	Work status.	-1	0: after-call processing. The business side is set to automatically enter the on-work state. 0–255: manual adjustment. An agent manually enters the on-work state by invoking the interface. The default value is 0, indicating the after-call processing state.
29	Forcibly sign out an agent.	Employee ID.	-1
30	Forcibly show an idle state for an agent.	Employee ID.	-1
31	Forcibly show a busy state for an agent.	Employee ID.	-1
32	Perform mute-on on a customer.	-1	-1
38	Hold a call by an agent.	Employee ID.	-1
40	Initiate an internal help request by an agent.	Employee ID.	0: Indicates no help between uncertain parties. 1: Indicates a two-party internal help. 2: Indicates a three-party internal help. 3: In speak transfer mode, the called party talks with the agent after answering the call; the called party is connected to the calling party when the agent releases the call. 4: In three-party conversation transfer mode, the three-party conversation is set up when the called party answers the call; the called party is connected to the calling party when the agent releases the call. 5: Indicates a three-party conversation. 6: Indicates that the call is to be connected.
41	Initiate an internal call by an agent.	Employee ID.	-1
43	Intercept a call.	Employee ID.	-1

Table 1 Agent operation type

Operation Type	Description	Operation Object	Operation Reason
44	Suspend voice playing.	-1	-1
45	Internal transfer.	Employee ID.	-1
46	Transfer-out.	-1	-1
47	Sign in to the media server.	-1	-1
52	Sign out of the media server.	-1	-1
53	Transfer a call that is queuing in a public queue.	-1	-1
54	Release transfer	-1	-1
55	Transfer to another device.	-1	-1
57	Hold the call when seeking internal help.	-1	-1
58	The agent enters the conference state.	Agent ID.	-1

5.5 Media Type

Table 1 Media type

Media Type	Description
1	Text chat
5	Common voice call

5.6 Description of Call Types

Call Type

Table 1 Call Type

ID	Call Type	Description
0	SP_CALL_NORMAL	Normal incoming call
5	SP_CALL_LONG	Incoming toll call
6	INTER_CALL	Internal call

Table 1 Call Type

ID	Call Type	Description
		It refers to a call made by an agent to another agent. An internal call cannot be forwarded, or muted.
7	SP_CALL_OUT	Normal outgoing call It refers to an outgoing call made by agents.
8	OP_CALL_OUT	Agent outgoing call type
9	OP_PRI_OUT	Outgoing call in the PRI mode
10	IVR_CALL_OUT	Outgoing call through IVR
11	SELF_CALLOUT	Outgoing call made by a Phone agent
12	IVR_PRI_CALL	Outgoing call in the PRI mode through IVR
13	NIRC_CALL_IN	Incoming call to a networked call center
14	NIRC_CALL_OUT	Outgoing call of a networked call center
15	NIRC_SPY_CALL_IN	QC incoming call to a networked call center
16	NIRC_SPY_CALL_OUT	QC outgoing call of a networked call center
17	NIRC_INTER_CALL_IN	Internal incoming network call to a virtual agent
18	NIRC_INTER_CALL_OUT	Internal outgoing network call of a virtual agent
20	ICD_SP_OPS_AIDERCALL	Aid call
21	ICD_SP_OPS_INCALL	Incoming call (international agent)
22	ICD_SP_OPS_PASSCALL	Relay call
23	ICD_SP_OPS_GIVECALL	Outgoing call (international agent)
40	OUTBOUND_CALL_OUT	Ordering an outgoing call The system queries the reserved outgoing call database regularly. After obtaining the reserved calls whose time is earlier than the current computer time, the system forcibly makes the calls.
41	ICD_SP_OUTBOUND_PRE_OCCUPY	Pre-occupied an outgoing call The system occupies an idle agent and then makes an outgoing call. If making the outgoing call succeeds, the idle agent handles the call.
42	ICD_SP_OUTBOUND_PRE_CONNECT	Pre-connecting an outgoing call The system makes an outgoing call to a user. If making the call succeeds, the system connects the call to an agent when the phone of the user rings.

Table 1 Call Type

ID	Call Type	Description
43	ICD_SP_OUTBOUND_VIRTUAL_CALLIN	Virtual incoming and outgoing calls The system makes an outgoing call to a user. If making the call succeeds, the system connects the call to an agent after the user hooks off.
44	ICD_SP_OUTBOUND_PREVIEW	Previewing an outgoing call It refers to a call that is made by an agent after the agent previews the outgoing call information.
45	ICD_SP_OUTBOUND_CALLBACK	Callback request If a Web user needs to contact a call center when browsing Web pages on Internet, the Web user can use the callback request service. An agent in the call center then dials the phone number specified by the Web user. After the call is connected, the web user follows the voice instruction of the agent and at the same time enjoy other services such as escorted browsing.
46	ICD_SP_IDD	International incoming toll call
50	ICD_SP_CBRT_CALL	RBT call
51	ICD_SP_INTERNAL_TWO_HELP	Two-party help call When answering an incoming call, an agent can ask for internal help. In the case of a two-party help, the call of the customer is held, and the agent talks with the asked-for-help agent.
52	ICD_SP_INTERNAL_THREE_HELP	Three-party help call When answering an incoming call, an agent can ask for internal help. In the case of a three-party help, the customer, agent, and asked-for-help agent can talk with each other. In the case of a two-party help, a three-party help can be initiated to form a three-party conversation.
60	ICD_SP_OUTBOUND_PRE_OCCUPY_PRI_CALL	Pre-occupied an outgoing call in the PRI mode
61	ICD_SP_OUTBOUND_PRE_CONNECT_PRI_CALL	Pre-connecting an outgoing call in the PRI mode
62	ICD_SP_OUTBOUND_VIRTUAL_CALLIN_PRI_CALL	Virtual incoming and outgoing calls in the PRI mode
63	ICD_SP_OUTBOUND_PREVIEW_PRI_CALL	Previewing an outgoing call in the PRI mode
64	ICD_SP_OUTBOUND_CALLBACK_PRI_CALL	Callback request in the PRI mode

5.7 Description of Call Types (Recording Index)

Table 1 Call types

No.	Call Type
0	Inbound call
1	Outbound call

5.8 Reasons for Leaving the Device of the Call

Table 1 Reasons for leaving the device of the call

Cause code	Description
0	Normal access/transfer.
1	Overflow transfer.
2	Timeout transfer.
3	Transfer on no agent.
4	Transfer on queue cancellation.
5	Transfer because the agent does not answer the call.
6	Resume a suspended call.
7	Connecting a held call.
8	Three-party help.
9	Picking a held call.
10	Three-party conversation.
11	The party requesting internal help releases the call.
12	The call is intercepted.
13	The call is picked up.
14	The analysis of the called is not configured.
15	The user releases the call.
16	The agent releases the call.
17	The call is transferred out.
18	The call is held.
19	An internal processing error occurs.

Table 1 Reasons for leaving the device of the call

Cause code	Description
20	The called party is busy.
22	The mobile phone is switched off.
23	The mobile phone is out of the service area.
24	The called number does not exist.
25	The ACD cannot distribute any resources for the call.
26	The user does not answer the call.
27	The called party rejects the call.
28	The call is successfully transferred to a skill queue.
29	The three-party conversation is transferred to a skill queue.
30	Transferring the call in the success transfer mode times out.
31	Transferring the three-party conversation to a skill queue times out.
34	If call timeout or queuing on busy or overflow occurs, the network call fails.
35	The network call is directed to its original call center for queuing.
36	The queued call is re-routed.
37	The call accesses the CDN through call routing.
38	The call accesses the CDN through agent consultation.
39	The CDN properly returns a routing result.
40	CDN routing times out.
41	An agent cancels the consultation call, or the routing fails.
42	The call accesses the PGR because CDN routing times out.
43	The call is properly returned by the CDN and is routed to the PGR.
44	The call accesses the PGR through inbound routing.
45	The call accesses the PGR through agent consultation.
46	The agent is unreachable. The CDN reroutes the call.
47	The call is waiting in multiple queues.
48	The primary-queue call is answered by numbers in the secondary queue.

Table 1 Reasons for leaving the device of the call

Cause code	Description
50	The agent rejects the call.
51	A user call joins in a conference call.
52	A conference chairperson proactively releases a conference, or the conference is released due to a CCS exception.
53	The participant is removed from the conference.
54	The call is transferred when the call completion rate is less than the threshold.
88	After the agent answers the call, the user hangs up.
89	After the agent answers the call, the user hangs up.

5.9 Description of Call Media Types

Media Type

Table 1 Media Type

ID	Media Type	Description
1	MEDIA_TYPE_CHAT	Text chat
2	MEDIA_TYPE_WEBPHONE	click-to-dial
3	MEDIA_TYPE_ESCORT	Escorted browsing and form sharing
4	MEDIA_TYPE_CALLBACK	Callback request call
5	MEDIA_TYPE_PHONE	Common voice call
6	MEDIA_TYPE_EMAIL	Email call
7	MEDIA_TYPE_FAX	Fax call
8	MEDIA_TYPE_VIDEO	IP video call (H.323)
9	MEDIA_TYPE_WB	Electronic white board
10	MEDIA_TYPE_APP_SHARE	Application sharing
11	MEDIA_TYPE_FILE_TRANSFER	File transfer
12	MEDIA_TYPE_VIDEO_2B1D	2B+D ISDN video call
13	MEDIA_TYPE_VIDEO_6B1D	6B+D ISDN video call

Table 1 Media Type

ID	Media Type	Description
14	MEDIA_TYPE_OPS	OPS call
15	MEDIA_TYPE_PREDICT_OUTBOUND	Predict dialing
16	MEDIA_TYPE_PREVIEW_OUTBOUND	Preview dialing
17	MEDIA_TYPE_MSG	Message media
18	MEDIA_TYPE_WEBPHONE_VIDEO	Video click-to-dial
19	MEDIA_TYPE_PHONE_VIDEO	Video call
20	MEDIA_TYPE_WEB_LEAVE_MESSAGE	Non-real-time call
21	MEDIA_TYPE_DESKTOP_SHARE	Desktop Sharing Media Type
22	MEDIA_TYPE_VC_CALL	Calling number in a skill queue
23	CLASSIC_MEDIA_TYPE_NUM	Total number of contact media types.
50	MEDIA_TYPE_MULTI_MEDIA_EMAILCHAT	E-mail chat
51	MEDIA_TYPE_MULTI_MEDIA_WEBCHAT	Internet Chat
52	MEDIA_TYPE_MULTI_MEDIA_SMSCHAT	SMS chat
53	MEDIA_TYPE_MULTI_MEDIA_SOCIALCHAT	Text chat
54	MEDIA_TYPE_MULTI_MEDIA_FAXCHAT	Fax chat
55	MEDIA_TYPE_MULTI_MEDIA_H5CHAT	H5 video

5.10 Description of Call Submedia Types

Table 1 Call submedia types (corresponding to call media type No. 53)

No.	Submedia Type	Description
1	SUB_MEDIA_TYPE_WEB	Web
2	SUB_MEDIA_TYPE_WHATSAPP	WhatsApp
3	SUB_MEDIA_TYPE_LINE	LINE
4	SUB_MEDIA_TYPE_WECHAT	WeChat
5	SUB_MEDIA_TYPE_FACEBOOK	FaceBook Messenger
6	SUB_MEDIA_TYPE_TWITTER	Twitter
7	SUB_MEDIA_TYPE_EAMIL	Email

Table 1 Call submedia types (corresponding to call media type No. 53)

No.	Submedia Type	Description
8	SUB_MEDIA_TYPE_SMS	SMS
9	SUB_MEDIA_TYPE_TELEGRAM	Telegram
10	SUB_MEDIA_TYPE_INSTAGRAM	Instagram